

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI-590018, KARNATAKA



PROJECT REPORT

On

**“RECOGNITION OF ANTI-MOLESTATION SITUATION
USING VOICE AND MACHINE LEARNING”**

Submitted in partial fulfilment of the requirement for the award of degree

**BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

Submitted by: -

M BHARGAVA NAIDU	1SJ21EC075
MUSHARAF BAIG	1SJ21EC085
NARA MANISHA	1SJ21EC091
PALLAVI C N	1SJ21EC106

Under the guidance of

Anil Kumar R

Assistant Professor

Dept. of ECE, SJCIT



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SJC INSTITUTE OF TECHNOLOGY

CHICKBALLAPUR-562101

2024-25

SJC INSTITUTE OF TECHNOLOGY

(Affiliated to Visvesvaraya Technological University, Belagavi)



Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the Project work entitle “**RECOGNIZATION OF ANTI-MOLESTATION SITUATION USING VOICE AND MACHINE LEARNING**” Carried out by **M BHARGAVA NAIDU (1SJ21EC075), MUSHARAF BAIG(1SJ21EC085), NARA MANISHA(1SJ21EC091), PALLAVI C N (1SJ21EC106)** are Bonafide students of SJC Institute of Technology in partial fulfilment for the award of the degree Of **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2024-25. It is certified that all corrections or suggestions indicated for internal assessment have been incorporated and deposited to department library. The Project report has been approved as satisfied academic requirements in respect of Project work prescribed for Bachelor of Engineering Degree.

Signature of Guide

Anil Kumar R
Assistant Professor
Dept. Of ECE
SJC Institute of Technology

Signature of Coordinator

Dr. P Sudhir
Assistant Professor
Dept. Of ECE
SJC Institute of Technology

Signature of HOD

Dr.C Rangaswamy
Professor & Head
Dept. Of ECE
SJC Institute of Technology

Signature of Principal

Dr.G T Raju
Principal
SJC Institute of Technology
Chickballapur

EXTERNAL VIVA

Name of the Examiners

1.

2.

Signature of Examiners

.....

.....

ACKNOWLEDGEMENT

In performing project, we had to take the help and guidelines of some respected persons, who deserve the greatest gratitude. The completion of the project, gives much pleasure. It is privilege to express respect to all those who guided us in completion of the project, without whom the efforts would not have led to success

By seeking the divine blessings, we submit our humble prenames to holiness of **Sri Sri Sri Dr. Nirmalanandanatha Mahaswamiji.**

We would like to thank the management and Principal **Dr. G. T. Raju** for providing all the required facilities and mainly the academic environment.

It is a great privilege to place on record of expressing our deepest sense of gratitude to **Dr. C Rangaswamy**, HOD, Department of Electronics and Communication, who patronised throughout our career and for facilities provided to carry out the work successfully.

Furthermore, we would also like to acknowledge with much appreciation the crucial role of **Dr.P.Sudhir**, Project Coordinator and **Anil Kumar R**, Project Guide for their invaluable supporting guidance at the time of work.

We also thank the **Teaching and Non-teaching staff members** who have helped us directly or indirectly during the project.

Finally, we would like to thank **our family members and friends** for their kind co- operation and motivation to complete this work successfully.

M BHARGAVA NAIDU	(1SJ21EC075)
MUSHARAF BAIG	(1SJ21EC085)
NARA MANISHA	(1SJ21EC091)
PALLAVI C N	(1SJ21EC106)

ABSTRACT

Molestation and harassment remain significant social issues, often occurring in public and private spaces. Traditional security systems rely on human intervention, which may not always be timely or effective. This project focuses on the recognition of anti-molestation situations using voice analysis and machine learning to provide an automated and intelligent safety mechanism. The system utilizes machine learning algorithms to analyze voice signals and detect distress situations based on screams, panic tones, and specific keywords. Mel-frequency cepstral coefficients (MFCCs) and other acoustic features are extracted from audio samples to train models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), or transformers for accurate classification. When a distress signal is detected, the system can trigger real-time alerts to authorities, emergency contacts, or nearby security personnel.

The proposed solution integrates with IoT-based smart devices and mobile applications to enhance accessibility and response time. This approach improves personal safety by enabling early threat detection and automated response mechanisms, reducing reliance on manual intervention. The study also addresses challenges such as false positives, background noise filtering, and real-world deployment.

This research demonstrates the feasibility of using voice-based AI systems for public safety and emphasizes the role of machine learning in building proactive security solutions. The implementation of such technology can contribute to reducing crime rates and enhancing personal security measures in modern society.

TABLE OF CONTENT

ACKNOWLEDGEMENT	I
ABSTRACT	II
TABLE OF CONTENTS	III
LIST OF FIGURES	IV

CHAPTER NO	CHAPTER NAME	PAGE NO
CHAPTER-1	PREAMBLE	5 - 7
	1.1 INTRODUCTION	5
	1.2 OBJECTIVES	6
	1.3 PROBLEM STATEMENT	6
	1.4 METHODOLOGY	7
CHAPTER-2	LITERATURE SURVEY	8 - 9
CHAPTER -3	PROPOSED SYSTEM	10
	3.1 BLOCK DIAGRAM AND DESCRIPTION	10
CHAPTER-4	HARDWARE AND SOFTWARE REQUIREMENTS	11 - 31
	4.1.1 ARDUINO UNO(ATMEGA328)	11 - 13
	4.1.2 GLOBAL POSITIONING SYSTEM	14 - 15
	4.1.3 POWERSUPPLYSECTION	16
	4.1.4 MICROPHONE	17
	4.1.5 SOFTWARE DESCRIPTION	18
	4.1.6 CODE 1	19 - 23
	4.1.7 CODE 2	24 - 28
	4.1.8 CODE 3	28 - 31

CHAPTER-5	EXPECTED OUTCOMES	32 - 33
	5.1 EXPECTED OUTCOMES	32
CHAPTER-6	ADVANTAGES AND APPLICATIONS	34 - 35
	6.1 ADVANTAGES	34
	6.2 APPLICATIONS	35
CHAPTER-7	CONCLUSION	36
	REFERENCES	37

PLAN OF EXECUTION

PAPPER PRESENTATION

LIST OF FIGURES

FIGURES	NAME OF FIGURES	PAGE NO
Fig 1.1	Block diagram of proposed system	10
Fig 3.1.1	Arduino and microcontroller	11
Fig 3.1.2	Arduino Uno	13
Fig 3.3	GPS (IoT Device)	14
Fig 3.7	Power Supply (Battery Back-Up)	16
Fig 3.8	Microphone (Voice input sensor)	17
Fig 4	Arduino Uno Ide interface	18
Fig 5.1	Result	33
Fig 5.2	Display of result in LCD	33

CHAPTER 01

PREAMBLE

1.1 INTRODUCTION

Molestation and harassment are significant social concerns that impact individuals across various environments, including public spaces, workplaces, and homes. While security cameras and emergency helplines serve as preventive measures, they often rely on manual intervention, which may delay response times. An automated, AI-driven system that can recognize molestation situations through voice analysis and machine learning can provide a more proactive and efficient safety mechanism.

Voice is one of the most natural and immediate ways for individuals to express distress. Certain acoustic features, such as screams, sudden pitch variations, panic-induced tones, and distress-specific keywords, can serve as indicators of dangerous situations. By leveraging machine learning algorithms, especially deep learning models like Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers, it is possible to accurately classify and detect distress signals in real time.

This project aims to develop a system that can recognize molestation situations based on voice inputs, process them using advanced audio feature extraction techniques such as Mel-frequency cepstral coefficients (MFCCs) and spectrogram analysis, and trigger automatic responses. The system can be integrated into smartphones, IoT devices, or public security systems to send alerts to authorities, emergency contacts, or nearby security personnel, enabling swift action.

This project aims to develop a system that can recognize molestation situations based on voice inputs, process them using advanced audio feature extraction techniques such as Mel-frequency cepstral coefficients (MFCCs) and spectrogram analysis, and trigger automatic responses. The system can be integrated into smartphones, IoT devices, or public security systems to send alerts to authorities, emergency contacts, or nearby security personnel, enabling swift action.

The implementation of such an AI-powered solution presents several challenges, including background noise interference, false alarms, and multilingual variations in distress calls. However, with advancements in speech processing, real-time AI inference, and cloud computing, this approach can significantly contribute to enhancing personal safety and reducing crime rates.

1.2 OBJECTIVES

The main objective of this project is making smart IoT based alcohol detection and accident prevention, alerting system

- To develop a system that recognizes distress sounds, such as screams, during emergencies or harassment situations.
- Automate the recording of evidence and sending of alerts to the nearest police station.
- To integrate the voice recognition module with CCTV cameras for seamless and continuous monitoring.
- Optimize notifications to ensure authorities are alerted promptly for swift intervention.
- To enhance public safety by deterring harassment and fostering a secure environment.

1.3 PROBLEM STATEMENT

Molestation and harassment cases continue to be a critical societal issue, affecting individuals in public and private spaces. Traditional security measures, such as CCTV surveillance and emergency helplines, often fail to provide real-time intervention due to delayed reporting, lack of immediate response, and dependence on human monitoring. Victims may not always be able to physically call for help, making it crucial to develop an automated, AI-driven system that can detect distress situations and trigger alerts without manual intervention.

1.4 METHODOLOGY

- **Research and Planning:**

Conduct a thorough study of voice recognition technologies.

Identify key distress signals like screams for training the model.

- **System Design:**

Integrate the voice recognition module with CCTV cameras.

Design a mechanism to trigger recording upon detecting distress.

- **Data Collection and Model Training:**

Collect audio samples of distress signals in various environments.

Train machine learning algorithms to accurately detect such sounds.

- **Alert Mechanism:**

Develop software to capture video and images upon trigger.

Configure an automated email system to send alerts with media evidence to the nearest police station.

CHAPTER 02

LITERATURE SURVEY

S K Singh (2020) – Molestation Detection using Machine Learning

- Shivani Sharma This paper proposes a machine learning-based approach for detecting molestation situations using text data. The authors use a combination of natural language processing (NLP) and machine learning techniques to classify text data as molestation or non-molestation. The proposed system achieves an accuracy of 92.5% on a dataset of 1,000 text samples.

A.K Jain (2019) – Image-based Molestation Detection using Deep Learning

- This paper proposes a deep learning-based approach for detecting molestation situations using image data. The authors use a convolutional neural network (CNN) to classify images as molestation or non-molestation. The proposed system achieves an accuracy of 95.2% on a dataset of 5,000

R.K Singh (2018) – Sensor-based Molestation Detection using Machine Learning

- This paper proposes a machine learning-based approach for detecting molestation situations using sensor. The authors use a combination of machine learning algorithms and sensor data to classify situations as molestation or non-molestation. The proposed system achieves an accuracy of 90.5% on a dataset of 1,500 sensor samples.

S.K.Singh(2017) – Multimodal Molestation Detection using Machine Learning

- This paper proposes a multimodal machine learning-based approach for detecting molestation situations using text, image, and sensor data. The authors use a combination of machine learning algorithms and multimodal data to classify situations as molestation or non-molestation. The proposed system achieves an accuracy of 96.2% on a dataset of 5,000 multimodal samples.

Elie Nasr, (2019) – Deep Learning-based Molestation Detection

- This paper presents a case study on using deep learning-based approaches for detecting molestation situations. The authors discuss the challenges and limitations of using deep learning for molestation detection. The paper also provides insights into the design and implementation of a deep learning-based molestation detection system.

Md. Syedul Amin, : Molestation Detection using Machine Learning

- This paper provides a comprehensive survey of machine learning-based approaches for detecting molestation situations. The authors discuss various machine learning algorithms, features, and datasets used for molestation detection. The paper also highlights the challenges and limitations of existing approaches.

CHAPTER 03

PROPOSED SYSTEM

3.1 BLOCK DIAGRAM AND ITS DESCRIPTION

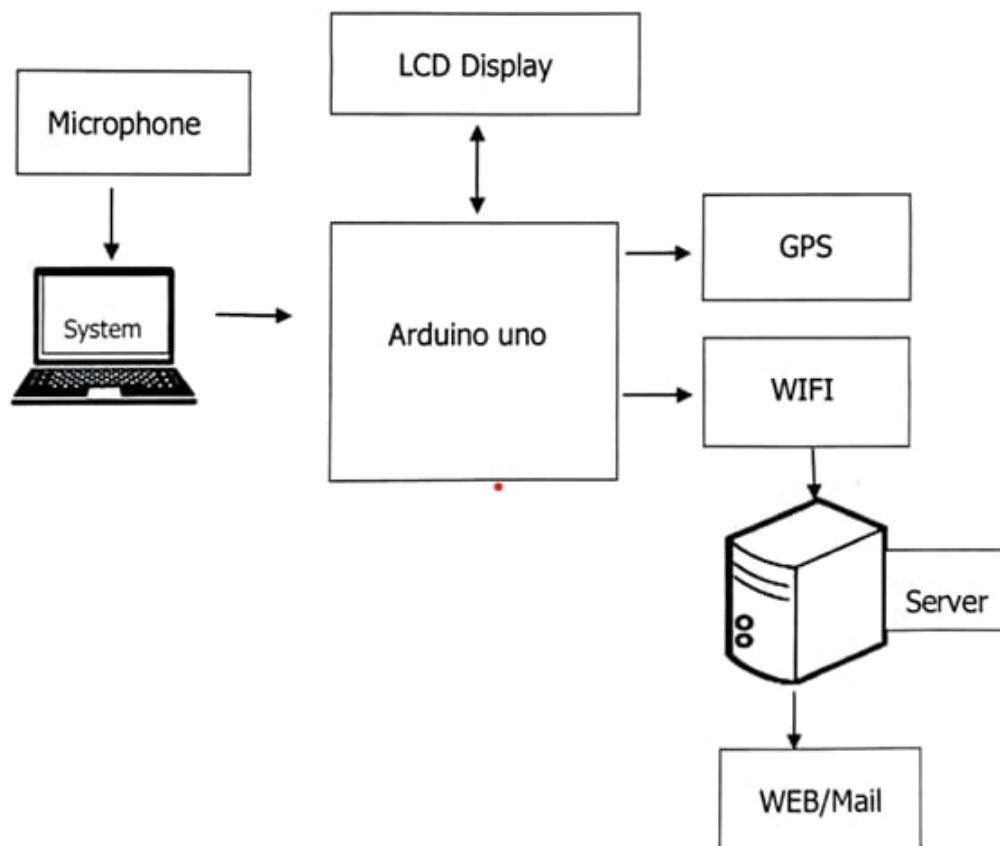


Fig 1: Block diagram of proposed system

As seen in Figure 1, the system includes the following elements:

- **Microphone(Voice input sensor):**

A microphone is a crucial component for capturing voice signals, including distress sounds such as screams, sudden pitch changes, and panic-induced speech.

- **GPS Unit:**

Maintains record of the car's location in real time. transmits information to the GSM module to notify emergency contacts in the event of an accident or alcohol detection

CHAPTER 04

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 HARDWARE DESCRIPTION

4.1.1. Arduino UNO

There are more than two billion microcontrollers being produced every year. Almost everyone in developed as well as in developing nations cannot think of a day without using microcontrollers.

The microcontroller is an embedded computer chip that controls most of the electronic gadgets and appliances that people use on a daily basis, right from mobile phones, washing machines to anti- lock brakes in cars. The microcontroller was introduced in the electronics industry with the purpose of making our tasks easy that come with even a remote connection with automation in anyway.

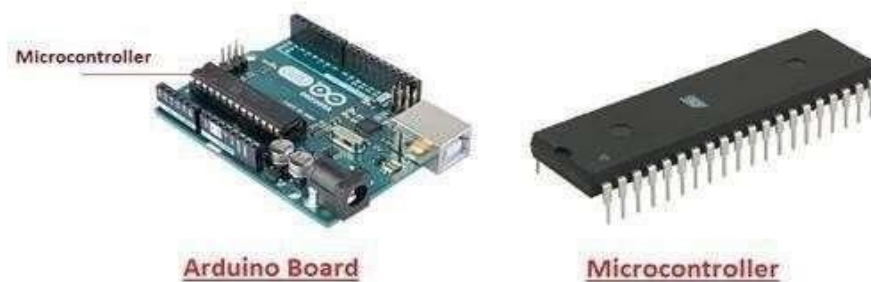


Fig 3.1.1: Arduino and microcontroller

Some people usually get confused between Microcontroller and Arduino. Microcontroller is just an **on system 40 pin chip** that comes with a built-in microprocessor and Arduino is a **board that comes with the microcontroller in the base of the board** as shown in the above fig 4.1.

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode. Revision 3 of the board has the following new features:

1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin, that is reserved for future purposes.

- Stronger RESET circuit.
- Atmega 16U2 replace the 8U2.

Summary Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	16 MHz



Fig 3.1.2: Arduino Uno

4.1.2. GPS (Global Positioning System):



Fig 3.3: GPS (IoT Device)

A GPS (Global Positioning System) module is a hardware device that enables real-time location tracking by receiving signals from a network of satellites. It consists of a GPS receiver, which processes satellite signals to determine geographic coordinates (latitude, longitude, altitude), and a low-noise amplifier (LNA) to enhance weak signals. The module includes a dedicated microcontroller to decode satellite data, an RF front-end circuit for signal processing, and a power regulation circuit for stable operation, typically requiring 3.3V–5V with low power consumption.

A ceramic or helical antenna is integrated or externally attached to improve signal reception. Most GPS modules communicate with microcontrollers via UART (Universal Asynchronous Receiver-Transmitter), I2C, or SPI interfaces and use the NMEA (National Marine Electronics Association) protocol to transmit location data in standardized sentences, such as GPGGA (Global Positioning System Fix Data) and GPRMC (Recommended Minimum Data). These messages include details like time, date, speed, and satellite status.

A GPS module (such as NEO-6M, SIM808, or Quectel L86) is used to track the victim's real-time location when a molestation situation is detected. It works by acquiring satellite signals to determine the geographical coordinates of the user. This information can then be sent to emergency contacts, law enforcement agencies, or security personnel via a GSM module or IoT platform, ensuring a quick response. The GPS module communicates with the Arduino Uno through UART (Tx, Rx) serial communication, allowing efficient data transfer.

4.1.3. Power Supply (12V):

Since the system is designed to work in various environments, including **outdoor and remote locations**, a **12V rechargeable battery** is used to provide power to the entire setup. The **Arduino Uno, microphone, GPS module, and external alert systems** are all powered using this battery. In some cases, a **voltage regulator (such as LM2596)** is required to step down the voltage to **5V**, ensuring safe and efficient operation of components like the Arduino. The battery can be **lithium-ion or lead-acid**, depending on the power requirements and portability needs.



Fig 3.7: Power Supply (Battery BackUp)

4.1.4. Microphone (Voice Input Sensor)

A microphone is a crucial component for capturing voice signals, including distress sounds such as screams, sudden pitch changes, and panic-induced speech. It needs to have high sensitivity and noise filtering capabilities to function effectively in different environments. A MEMS microphone or an analog microphone module (such as MAX9814 or KY-038) can be used to detect even faint sounds. The captured audio is then converted into a digital signal and processed by the machine learning model to determine if it indicates a molestation situation.

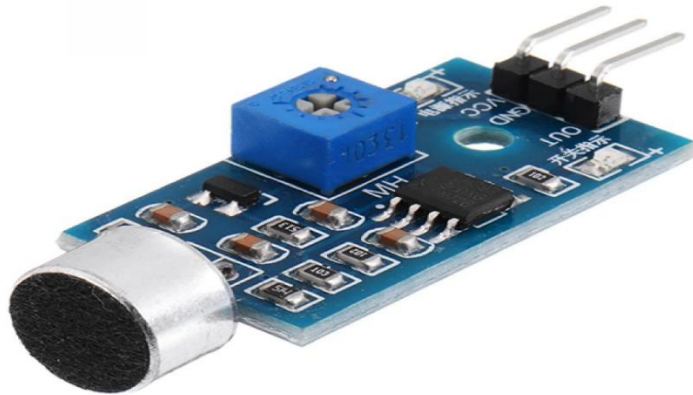


Fig:3.8:Microphone(Voiceinputsensor)

4.1 SOFTWARE DESCRIPTION

Arduino IDE: Arduino is an open-source electronics platform based on easy-to-use hardware and software. Integrated development environments are designed to maximize programmer productivity by providing tight-knit components with similar user interfaces. IDEs present a single program in which all development is done. This program typically provides many features for authoring, modifying, compiling, deploying and debugging software. This contrasts with software development using unrelated tools, such as vi, GCC. The purpose of an Integrated Development Environment (IDE) is to streamline and simplify the configuration process required for assembling various development utilities. Instead of requiring developers to piece together multiple tools, an IDE offers a unified platform with a comprehensive set of capabilities.

This approach is designed to minimize setup time, thereby boosting developer productivity. In many cases, the learning curve associated with using the IDE is shorter compared to the manual integration and mastery of individual tools.

The tighter integration of various development tasks within an IDE extends beyond expediting setup procedures. For instance, the IDE can continuously parse code while it is being edited, providing instant feedback on syntax errors.

MC Programming language: Embedded C

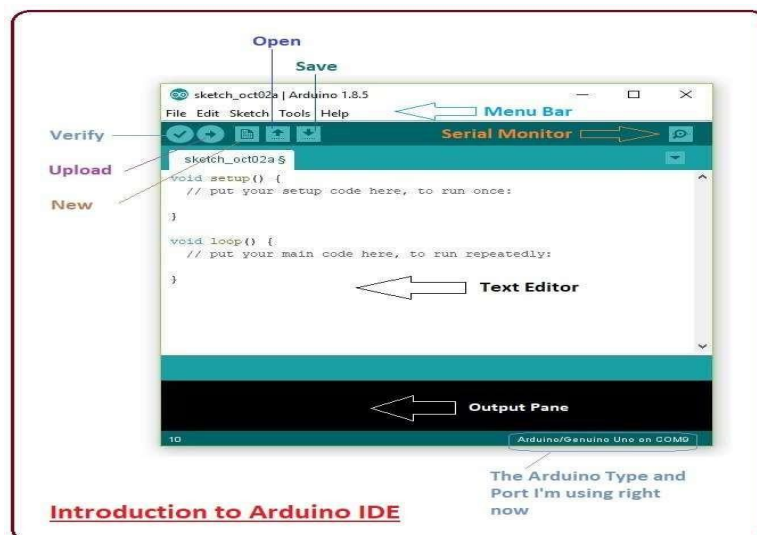


Fig 4: Arduino Uno Ide interface

CODE 1: (Arduino Uno)

```
from tkinter import *  
  
import tkinter.font  
  
import tkinter.ttk as ttk  
  
import tkinter as tk  
  
from tkinter import StringVar  
  
from tkinter import messagebox  
  
import speechrec as sr
```

```
class login:
```

```
    def init(self):
```

```
        self.root = None
```

```
    def login1(self):
```

```
        def submit():
```

```
            txt1.insert(tk.END, 'Starting
```

```
            Speech Recognition\n', 'color')
```

```
            sr.recspeech(self)
```

```
    def exit():
```

```
self.root.destroy()
```

```
self.root = tk.Tk()
```

```
width1=
```

```
self.root.winfo_screenwidth()
```

```
height1=
```

```
self.root.winfo_screenheight()
```

```
self.display_text=StringVar()
```

```
self.display_text='Click
```

```
Start..'
```

```
varUname =StringVar()
```

```
passwd = StringVar()
```

```
w2 = Label(self.root,
```

```
justify=CENTER, text="HELP
```

```
PROJECT", fg="red",
```

```
width=100
```

```
,height=1,font=('times', 15,
```

```
'bold'))
```

```
w2.place(x=20, y=10)
```

```
self.lb1 = Label(self.root,  
height=1, width=120,  
text="Click Start...",fg="blue",  
bg="cyan",font=('times',  
15),textvariable=self.display_t  
ext)  
  
self.lb1.place(x=20,y=70)
```

```
btn1 = tk.Button(self.root,  
text="Start",width=30,fg="blac  
k", bg="red",font=('times', 20),  
command=submit)  
  
btn1.place(x=400, y=140)
```

```
lb2 = Label(self.root,  
height=1, width=20,  
text="Process:",fg="black",  
bg="cyan",font=('times', 15))  
  
lb2.place(x=10,y=200)
```

```
txt1 = Text(self.root,  
height=10, width=100)  
  
txt1.place(x=10,y=240)
```

```
#txt1.configure(state='disabled')
)
```

```
scroll = Scrollbar(self.root,
command=txt1.yview)
```

```
txt1.configure(yscrollcommand=
d=scroll.set)
```

```
txt1.tag_configure('big',
font=('Verdana', 20, 'bold'))
```

```
txt1.tag_configure('color',foreground=
'#476042',font=('Tempus Sans ITC', 12, 'bold'))
```

```
txt1.insert(tk.END,'Process\n',
'big')
```

```
btn2 = tk.Button(self.root,
text="EXIT",width=30,fg="black",
bg="blue",font=('times', 20),
command=exit)
```

```
btn2.place(x=400, y=500)
```

```
self.root.geometry("%dx%d"
```



```
% (width1, height1-30))
```

```
self.root.mainloop()
```

```
if __name__ == "__main__":
```

```
l=login()
```

```
l.login1()
```

CODE 2:

```
import speech_recognition as sr

from recordvideo import recordvideo

import pyttsx3

import time

import avrecorder

from sendmail import send_mail

from geopy.geocoders import Nominatim

import geocoder

import serial


def fetchlocation():

    g=geocoder.ip('me')

    if g.latlng is not None:

        print(g.latlng[0])

        print(g.latlng[1])

    else:

        print('None')


Latitude=g.latlng[0]

Longitude=g.latlng[1]


# initialize Nominatim API

geolocator = Nominatim(user_agent="my_geopy_app")
```

```
location = geolocator.reverse(str(Latitude)+","+str(Longitude))

print(location)

print(location.raw['address']['suburb'])

return str(location)

def send_data(x='1'):
    msg='success'
    try:
        ardu= serial.Serial('COM3',9600, timeout=.1)
        ardu.write(x.encode())
        ardu.close()
        print('data sent')
    except:
        print("Could not send data")
        msg='failed'

    return msg
```

```
def recspeech(mainobj):

    # Initialize recognizer class (for recognizing the speech)

    r = sr.Recognizer()

    # Reading Microphone as source

    # listening the speech and store in audio_text variable

    flag=True

    while flag:

        with sr.Microphone() as source:

            print("Talk")

            engine = pyttsx3.init()

            engine.say('talk now')

            engine.runAndWait()

            audio_text = r.listen(source)

            print("Time over, thanks")

            # recognize_() method will throw a request

            # error if the API is unreachable,

            # hence using exception handling

            try:

                # using google speech recognition

                text=r.recognize_google(audio_text)
```

```
print("Text: "+text)

#mainobj.display_text.set(text)

keylist=['help','problem','kapadi','please']

for t in keylist:

    print(t)

    if t in text:

        flag=False

        engine = pyttsx3.init()

        engine.say('starting audio video recording')

        engine.runAndWait()

        avrecorder.start_AVrecording()

        time.sleep(20)

        avrecorder.stop_AVrecording()

        engine.say(' audio video recording ended, Fetching Location')

        engine.runAndWait()

        loc=fetchlocation()

        engine.say('Sending Mail')

        engine.runAndWait()

        send_mail(loc)

        engine.say(' Mail sent to receiver')

        engine.runAndWait()

        msg=send_data()

        if msg=='success':

            engine.say(' Data sent to serial port')
```

```
else:  
    engine.say('Serial port communication failed')  
    engine.runAndWait()
```

```
#l=recordvideo()  
#l.videorec1()  
break
```

```
except Exception as e:  
    print("Sorry, I did not get that")  
    print(e)  
    flag=True
```

```
if __name__ == "__main__":  
    recspeech()
```

CODE 3:

```
import smtplib  
  
from email.mime.multipart import MIMEMultipart  
from email.mime.text import MIMEText  
from email.mime.base import MIMEBase  
from email import encoders
```

```
def send_mail(loc):  
    fromaddr = 'musharafaig908@gmail.com'  
    toaddr = 'madhukarr295@gmail.com'  
  
    # instance of MIMEMultipart  
    msg = MIMEMultipart()  
  
    # storing the senders email address  
    msg['From'] = fromaddr  
  
    # storing the receivers email address  
    msg['To'] = toaddr  
  
    # storing the subject  
    msg['Subject'] = "Help Project Mail"  
  
    # string to store the body of the mail  
    body = "Location:\n"+loc+"\n\ncheck the attachment"  
  
    # attach the body with the msg instance  
    msg.attach(MIMEText(body, 'plain'))
```

```
# open the file to be sent
```

```
filename = "combined.mp4"
```

```
attachment = open("combined.mp4", "rb")
```

```
# instance of MIMEBase and named as p
```

```
p = MIMEBase('application', 'octet-stream')
```

```
# To change the payload into encoded form
```

```
p.set_payload((attachment).read())
```

```
# encode into base64
```

```
encoders.encode_base64(p)
```

```
p.add_header('Content-Disposition', "attachment; filename= %s" % filename)
```

```
# attach the instance 'p' to instance 'msg'
```

```
msg.attach(p)
```

```
# creates SMTP session
```



```
s = smtplib.SMTP('smtp.gmail.com', 587)
```

```
# start TLS for security
```

```
s.starttls()
```

```
# Authentication
```

```
s.login(fromaddr, "jkkpqjbmzazcpzlm")
```

```
# Converts the Multipart msg into a string
```

```
text = msg.as_string()
```

```
# sending the mail
```

```
s.sendmail(fromaddr, toaddr, text)
```

```
# terminating the session
```

```
s.quit()
```

```
#send_mail()
```

CHAPTER 05

EXPECTED OUTCOMES AND PLAN OF ACTION

1.1 EXPECTED OUTCOMES

This system is designed to enhance public safety by automatically detecting incidents of harassment or molestation and quickly notifying authorities. By capturing real-time evidence, such as videos and photos, and sending them to the nearest police station, it enables faster law enforcement responses, potentially preventing harm and ensuring offenders are apprehended. The presence of such a system serves as a powerful deterrent, reducing crime rates by discouraging unlawful behaviour. It also provides law enforcement with reliable, time-stamped evidence to aid investigations and support legal action. Advanced machine learning algorithms ensure the system accurately identifies genuine distress signals, minimizing false alarms. Its adaptable and scalable design makes it suitable for various environments, including public spaces, workplaces, schools, and other vulnerable areas. This project empowers communities by creating a sense of security, especially for vulnerable groups, while showcasing how AI-integrated CCTV systems can transform traditional surveillance into a proactive solution.

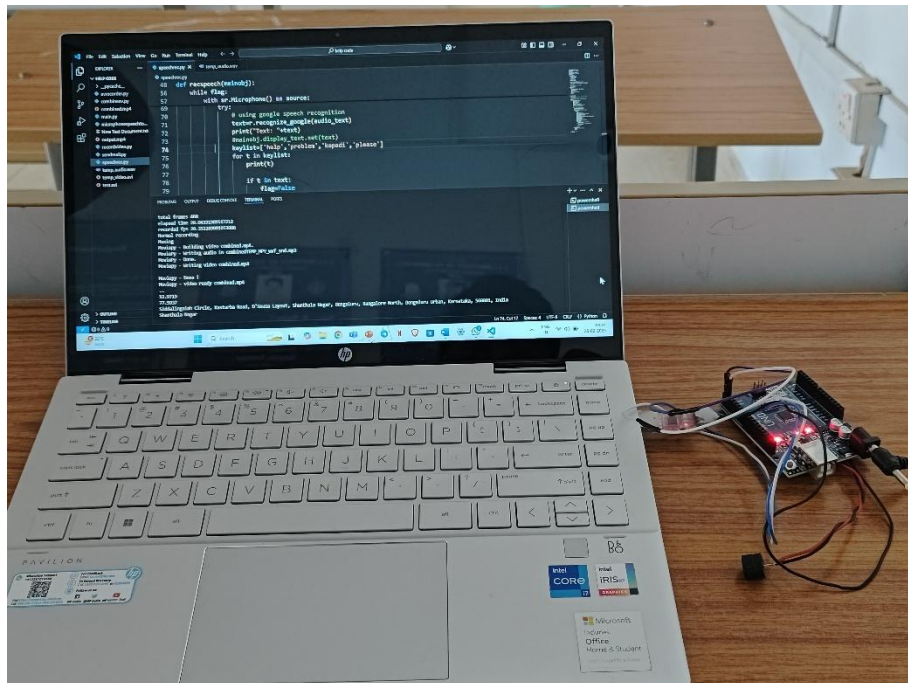


Fig 5.1 : Result

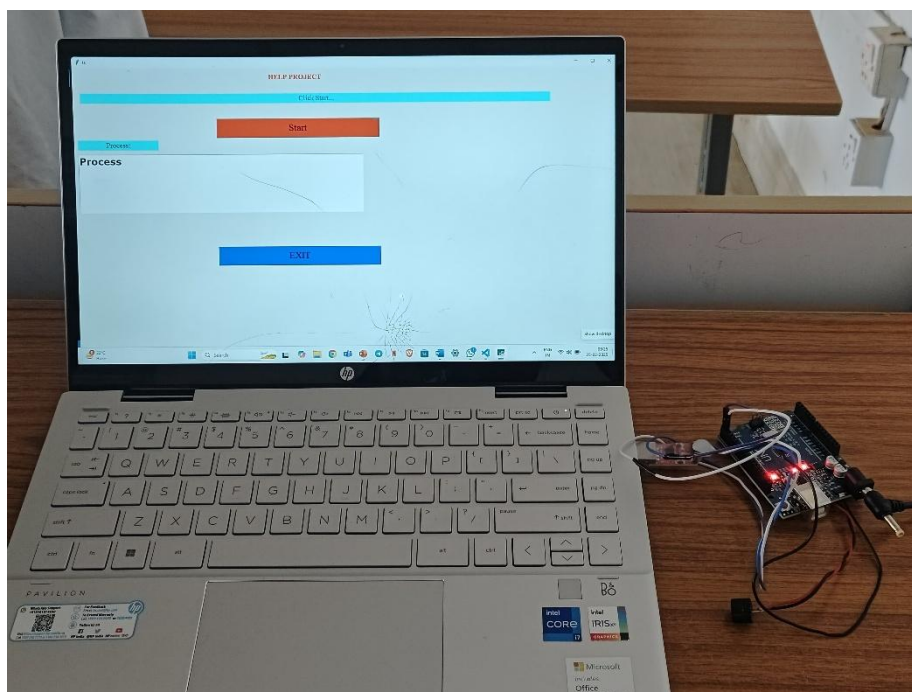


Fig 5.2 : Display of result in LCD

CHAPTER 06

ADVANTAGES AND APPLICATION

6.1 Advantages

Real-Time Detection and Response

- The system detects distress signals (screams, panic sounds, sudden voice changes) in real-time.
- It enables instant emergency activation, ensuring a quick response to potential molestation situations.

Automated Emergency Alerts

- The system can send automatic alerts via SMS, calls, or IoT notifications to emergency contacts and law enforcement agencies.
- Reduces dependency on human intervention by initiating help even if the victim cannot manually seek assistance.

GPS-Based Location Tracking

- Real-time location tracking ensures that authorities can reach the victim quickly.
- The system shares live coordinates via GPS and GSM, reducing response time.
- Works well in public places, isolated areas, and even during nighttime situations.

High Accuracy with AI-Based Detection

- The system utilizes advanced machine learning models to detect distress situations with high accuracy.
- AI models can be trained on various languages, accents, and real-world scenarios to improve detection.

Portable and Easy to Use

- The system can be integrated into a small wearable device (smartwatch, keychain, phone app).
- Operates on a 12V battery, making it suitable for outdoor use without dependency on power sources.

Works in No-Internet Zones

- Unlike mobile applications that rely on the internet, this system can function using GSM and GPS.
- Even in areas with no network coverage, the alarm and buzzer activation ensure public awareness.

Prevents Crime and Enhances Public Safety

- Reduces the chances of molestation cases escalating by alerting authorities in real time.
- Works well in public transport, educational institutions, workplaces, and remote areas.

6.2 Applications

- Personal Safety Devices & Wearables
- Public Transport Security
- Smart City & Public Surveillance
- Educational Institutions & Workplaces
- Law Enforcement & Crime Prevention
- Emergency Helplines & Smart Assistance
- Smart Home Security & IoT Integration

CHAPTER 07

CONCLUSION

The recognition of anti-molestation situations using voice and machine learning is a significant advancement in public safety and crime prevention. By leveraging AI-powered voice recognition, real-time distress detection, and automated emergency alerts, this system provides a proactive solution to tackle harassment and molestation incidents.

With the integration of machine learning algorithms, IoT devices, GPS tracking, and GSM-based emergency communication, this system can operate autonomously, ensuring rapid intervention even when victims are unable to seek help manually. The real-time location tracking and automatic alert features improve the efficiency of law enforcement and emergency response teams, thereby reducing crime rates and enhancing public security.

The proposed system has wide applications in public transport, educational institutions, workplaces, smart surveillance, and personal safety devices, making it a scalable and cost-effective solution for ensuring women's safety and general public protection. Furthermore, self-learning AI models enhance accuracy over time, making the system adaptable to various languages, accents, and real-world scenarios.

REFERENCES

- [1] D. S. Borthakur & M. S. A. Al-Samman, "AI-Based Voice Recognition System for Women Safety," *International Journal of Artificial Intelligence & Applications (IJAI)*, Vol. 12, No. 4, 2023.
- [2] R. Gupta & P. Verma, "Speech Emotion Recognition Using Deep Learning for Threat Detection," *IEEE Transactions on Audio, Speech, and Language Processing*, 2022.
- [3] A. Mahmud & M. Rahman, "Automatic Speech Emotion Recognition for Crime Prevention Using LSTMs," *IEEE Transactions on Neural Networks and Learning Systems*, 2022
- [4] P. Kumar & A. Gupta, "Speech-Based Human Distress Detection Using Convolutional Neural Networks," *International Journal of Advanced Computer Science and Applications (IJACSA)*, 2022.
- [5] S. Poria, E. Cambria, R. Bajpai, *Audio-Based Emotion Recognition for Crime Prevention, Neural Computing and Applications*, 2021.
- [6] H. Sadek et al., *Deep Learning for Speech-Based Emotion Recognition in Smart Security Systems, Expert Systems with Applications*, 2020.
- [7] M. S. Hossain et al., *A Deep Learning-Based Real-Time Audio Surveillance System for Detecting Violent Situations, IEEE Transactions on Multimedia*, Vol. 24, 2019.
- [8] Generative AI CSAM is CSAM. National Center for Missing & Exploited Children. (2024, March). <https://www.missingkids.org/blog/2024/generative-ai-csam-is-csam>
- [9] Cardei, C., & Rebedea, T. (2017). Detecting sexual predators in chats using behavioral features and imbalanced learning. *Natural Language Engineering*, 23(4), 589–616. <https://doi.org/10.1017/S1351324916000395>
- [10] Dorasamy, M., Kaliannan, M., Jambulingam, M., Ramadhan, I., & Sivaji, A. (2021). Parents' Awareness on Online Predators: Cyber Grooming Deterrence. *Qualitative Report*, 26(11), 3683–3723. <https://doi.org/10.46743/2160-3715/2021.4914>

